

A 2-adic point on the five-row comparison variety

Exact computational certificate; independent review requested

30 September 2026

Proposition 1. *The five-row, four-column centered comparison moment equations have a solution over $\mathbb{Q}_2$. Thus 2-adic local insolubility cannot exclude rational five-row models.*

Proof. Write $\mathbb{E}$ for averaging five coordinates. Set

$$r = 3/4, \quad s = 1/4, \quad p = C/16, \quad q = 5/6 - p, \quad R = p/r, \quad S = q/s,$$

and put

$$D = r^2q - s^2p, \quad z_1 = \frac{5r(q - s^2)}{6D}, \quad z_2 = \frac{5s(r^2 - p)}{6D}.$$

Let

$$z = (-z_1, -z_2, 0, z_2, z_1), \quad u = (q, -p, 2(p - q), -p, q).$$

We seek the four columns

$$a = z + \lambda u, \quad b = z - \lambda u, \quad c = (-r, -s, 0, s, r), \quad d = (-R, -S, 0, S, R).$$

The means vanish. Direct substitution gives $\mathbb{E}cd = \mathbb{E}zc = \mathbb{E}zd = 1/3$ and $\mathbb{E}ucd = 0$. Oddness under row reversal then verifies all four triple moments and the other mixed pair moments. The pair ab has the correct moment when

$$\lambda^2 = L(C) := \frac{64(C^2 - 24C + 148)}{3(C - 12)^2(3C^2 - 40C + 160)}.$$

Under this condition the remaining fourth moment $\mathbb{E}abcd = 1/5$ is equivalent, whenever the displayed denominators are nonzero, to

$$f(C) := 36C^5 - 1647C^4 + 29612C^3 - 264376C^2 + 1193920C - 2229120 = 0.$$

All these identities can also be checked by clearing their rational polynomial denominators.

Take $C_0 = 2179188$. Exact integer evaluations give

$$v_2(f(C_0)) = 28, \quad v_2(f'(C_0)) = 6.$$

Hensel's lemma therefore gives a root $C_* \in \mathbb{Z}_2$ with $C_* - C_0 \in 2^{22}\mathbb{Z}_2$. For the displayed integer numerator $P(C)$ and denominator $Q(C)$ of $L(C)$,

$$v_2(P(C_0)) = 8, \quad v_2(Q(C_0)) = 10, \quad 4L(C_0) \equiv 1 \pmod{8}.$$

Integer polynomials change by a multiple of 2^{22} throughout this ball. It follows that $v_2(L(C_*)) = -2$ and $4L(C_*) \equiv 1 \pmod{8}$. The usual square criterion in $\mathbb{Q}_2$ now supplies a square root $\lambda \in \mathbb{Q}_2$. The same valuation calculation keeps $Q(C_*)$ nonzero. Since

$$D = -5(C - 12)/128, \quad \mathbb{E}u^2 = 5/6 - 4pq = (3C^2 - 40C + 160)/192,$$

all denominators needed in the construction are nonzero. The four columns therefore satisfy all fifteen required moment equations. $\square$

This is a local-solubility result only. It proves neither a rational model nor a monotone real model, and makes no claim of realizability by actual dice. An elementary separate argument shows that the centered coordinates of any such model cannot all lie in $\mathbb{Z}_2$; the point here indeed has negative valuations.